Egypt-Japan University of Science and Technology
الجامعة المصرية اليابانية للعلوم و التكنولوجيا
エジプト日本科学技術大学

EECE
SCHOOL OF ENERGY RESOURCES, ENVIRONMENT, CHEMICAL AND PETROCHEMICAL ENGINEERING

Intro. Energy, Environmental and Chemical Eng. (CPE 211)

Assignment#2 "Environmental Engineering"

Question#1 (14 marks)

For certain rural areas in the country, vehicles are used to transport a yearly traffic volume of 3×10^7 by petrol. In the future, the yearly traffic volume will be increased by twofold, and therefore, vehicles will be replaced by buses that utilize natural gas for transportation. It's required to answer the following:

1.	The parameter a_m,CO_2 for vehicles= … kg/MJ							
	(A)	0.0369	(B)	0.0561	(C)	0.0693	(D)	0.0741
2.	The parameter a_m,CH_4 for vehicles= … kg/MJ							
	(A)	0.3×10^{-3}	(B)	0.008×10^{-3}	(C)	0.0974×10^{-3}	(D)	None of them
3.	The parameter a_m,N_2O for vehicles= … kg/MJ							
	(A)	0.002×10^{-3}	(B)	0.009×10^{-3}	(C)	0.022×10^{-3}	(D)	None of them
4.	The parameter b_m for vehicles= … MJ/veh./km							
	(A)	2	(B)	6	(C)	2.9	(D)	3.6
5.	The parameter C_m for vehicles= … Pass/veh.							
	(A)	200	(B)	30	(C)	2	(D)	300
6.	The parameter a_m,CO_2 for busses= … kg/MJ							
	(A)	0.0369	(B)	0.0693	(C)	0.0741	(D)	0.0561
7.	The parameter a_m,CH_4 for busses= … kg/MJ							
	(A)	0.0974×10^{-3}	(B)	0.3×10^{-3}	(C)	0.008×10^{-3}	(D)	None of them
8.	The parameter a_m,N_2O for busses= … kg/MJ							
	(A)	0.009×10^{-3}	(B)	0.022×10^{-3}	(C)	0.002×10^{-3}	(D)	None of them
9.	The parameter b_m for busses= … MJ/veh./km							
	(A)	20	(B)	78.9	(C)	14	(D)	6
10.	The parameter C_m for buses= … Pass/veh.							
	(A)	3	(B)	30	(C)	300	(D)	None of them
11.	The currently Q_{total} from the vehicles= … $\times10^6$ kg CO_2/year							
	(A)	9.27	(B)	6.54	(C)	2.49	(D)	5.39
12.	The expected Q_{total} in the future from the vehicles= … $\times10^6$ kg CO_2/year							
	(A)	13.10	(B)	5.00	(C)	18.54	(D)	8.66
13.	The expected Q_{total} in the future from the busses= … $\times10^6$ kg CO_2/year							
	(A)	0.24	(B)	5.44	(C)	4.12	(D)	7.32
14.	The percentage of CO_2 pollution reduction when buses will be used= … %							
	(A)	86.4	(B)	39.6	(C)	98.7	(D)	60.2

Parameter	Value
a_{m,CO_2} (Kg/MJ)	Electricity = 0.0369 Diesel = 0.0741 Petrol = 0.0693 Natural gas = 0.0561
b_m MJ/veh/Km	Petrol vehicles = 3.6 Diesel vehicles =2.9 Tram = 20 Train = 78.9 Bus or truck=14 Natural gas: Vehicle = 2 Bus = 6
C_m Pass/veh Ton/veh	Car = 2 Taxi = 3 Bus = 200 Tram = 300 Subway or train = 200 Truck = 30
a_{m,CH_4} (Kg/MJ)	Diesel vehicles=0.008×10^{-3} Petrol vehicles= 0.0974×10^{-3} Natural gas= 0.3×10^{-3}
a_{m,N_2O} (Kg/MJ)	Diesel vehicles=0.009×10^{-3} Petrol vehicles= 0.022×10^{-3} Natural gas= 0.002×10^{-3}
$e_{CH_4} = 11$	$e_{N_2O} = 270$

$$q_{m,CO_2} = \frac{a_{m,CO_2} \times b_m \times F_1 \times F_2}{C_m}$$

$$q_{m,N_2O} = \frac{a_{m,N_2O} \times b_m \times F_1 \times F_2}{C_m} \times e_{N_2O}$$

$$q_{m,CH_4} = \frac{a_{m,CH_4} \times b_m \times F_1 \times F_2}{C_m} \times e_{CH_4}$$

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Submit before Tuesday (29 Nov. 2022) 11:59 PM